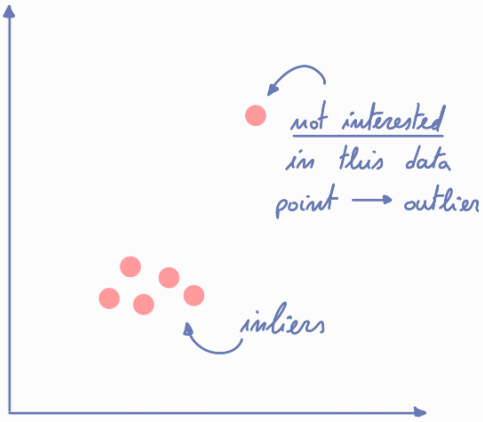
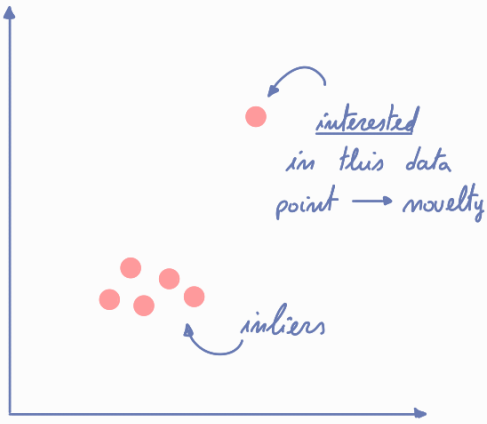
Novelty vs Outlier



Outlier example

Example: measuring ph of water; we know that usually the ph falls in some range and if we get a completely different value it will probably be due to a measurement error.



Novelty example

Example: credit card transactions; most of the transactions are good, some of them are frauds and we are interested in them

We have various methods to deal with this

- Robust covariance
- One-class SVM
- Isolation Forest
- Local Outlier Factor